RS Vorlesung 3

HA 1)

$$(2+2s)I_1 - (1+2s)I_2 - I_3 = 0$$

$$-(1+2s)I_1 + (7+5s)I_2 - (2+3s)I_3 = U(s)$$

$$-I_1 - (2+3s)I_2 + \left(3+3s+\frac{1}{5s}\right)I_3 = 0$$

$$\begin{pmatrix} (2+2s) & -(1+2s) & -1 \\ -(1+2s) & (7+5s) & -(2+3s) \\ -1 & -(2+3s) & \left(3+3s+\frac{1}{5s}\right) \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \\ I_3 \end{pmatrix} = \begin{pmatrix} 0 \\ U(s) \\ 0 \end{pmatrix}$$

$\Rightarrow$ MatLab

Vorlesung_3_Hausaufgaben.mlx

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clear all;
close all;
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Hausaufgabe 1

```
syms s U(s) Uo(s) I1(s) I2(s) I3(s)

% Definition
I = [I1(s); I2(s); I3(s)];
A = [2+2*s     -(1+2*s)    -1
    -(1+2*s)  7+5*s       -(2+3*s)
    -1        -(2+3*s)     3+3*s+1/(5*s)]
U = [0; U(s); 0];

% Berechnung
I = inv(A)*U
Uo = I(2)*4;
G(s) = Uo/U(2);

% Ausgabe
disp('Uo/U =');
disp(G(s));
```

A =

$$\begin{pmatrix} 2s+2 & -2s-1 & -1 \\ -2s-1 & 5s+7 & -3s-2 \\ -1 & -3s-2 & 3s+\frac{1}{5s}+3 \end{pmatrix}$$

I =

$$\begin{pmatrix} \frac{U(s)\ (30s^3+60s^2+27s+1)}{\sigma_1} \\ \frac{U(s)\ (30s^3+60s^2+27s+2)}{\sigma_1} \\ \frac{5s\,U(s)\ (6s^2+12s+5)}{\sigma_1} \end{pmatrix}$$

where

$$\sigma_1 = 120s^3+246s^2+120s+13$$

Uo/U =

$$\frac{4\ (30s^3+60s^2+27s+2)}{120s^3+246s^2+120s+13}$$

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HA 3)

$$\begin{pmatrix} 1+2s+s^2 & -(1+s) \\ -(1+s) & 1+s+s^2 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} F(s) \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \frac{1}{s^4+3s^3+2s^2+\cancel{2}s^2+3s+1-s^2-2s-1} \begin{pmatrix} 1+s+s^2 & 1+s \\ 1+s & 1+2s+s^2 \end{pmatrix} \begin{pmatrix} F(s) \\ 0 \end{pmatrix}$$

$$\frac{X_2}{F(s)} = \frac{1+s+s^2}{s^4+3s^3+s^2+s}$$

HA 2) 1)

$$T_1\, \dot{y}(t) + y(t) = K\, u(t)$$

$$\circ\!\!-\!\!\bullet$$

$$T_1\, Y(s)\, s + Y(s) = K\, U(s)$$

$$\frac{Y(s)}{U(s)} = \frac{K}{T_1 s+1} = \frac{\frac{K}{T_1}}{s+\frac{1}{T_1}} \quad \bullet\!\!-\!\!\circ \quad \frac{K}{T_1}\, e^{-\left(\frac{1}{T_1}\right)t}$$

3)

$$\frac{Y(s)}{U(s)} = \frac{K}{T_1 s^2 + s}$$

2)

$$\ddot{y}(t) + 2D\omega_0\, \dot{y}(t) + \omega_0^2\, y(t) = K\, u(t)$$

$$\circ\!\!-\!\!\bullet$$

$$\frac{Y(s)}{U(s)} = \frac{K}{s^2+2D\omega_0 s+\omega_0^2} \quad \bullet\!\!\not\!\!-\!\!\circ$$

$$\cancel{\frac{K}{(s+\omega_0)^2 + Ds}}$$

$D-1$

4)

$$\frac{Y(s)}{U(s)} = \frac{K+T_D s}{T_1 s^2 + T_2 s + 1}$$